

## WORKBOOK ASSIGNMENT 7

### Questions:

Q1) Create a MySQL database named CollegeDB and create two tables: Course and StudentResult.

### ANS

```
1 CREATE DATABASE CollegeDB;|
2
3
4
```

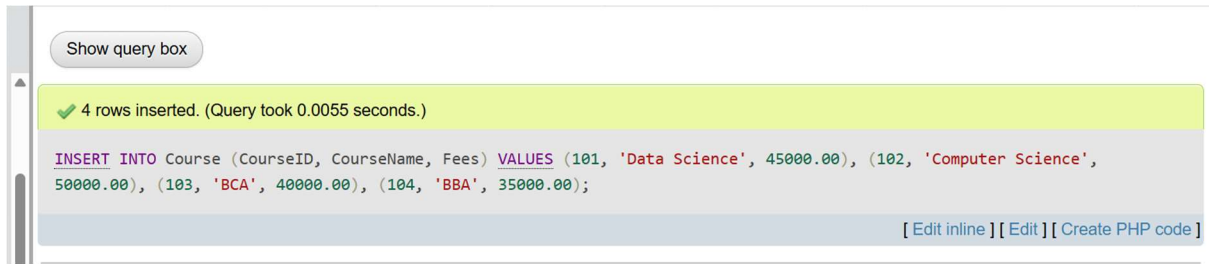
```
1 USE CollegeDB;|
2
3
4
5
```

```
1 CREATE TABLE Course (
2     CourseID INT PRIMARY KEY,
3     CourseName VARCHAR(50) NOT NULL,
4     Fees DECIMAL(10,2) NOT NULL
5 );
6
7
```

```
1 CREATE TABLE Student (
2     StudentID INT PRIMARY KEY,
3     StudentName VARCHAR(50) NOT NULL,
4     Age INT,
5     City VARCHAR(30),
6     CourseID INT,
7     FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
8 );|
```

Q2) Write a MySQL query to display students whose marks are greater than the average marks of all students.

Ans



MySQL query execution interface showing a successful INSERT INTO statement. The query inserted 4 rows into the Course table. The interface includes a 'Show query box' button, a green success message, the SQL code, and links for 'Edit inline', 'Edit', and 'Create PHP code'.

```
INSERT INTO Course (CourseID, CourseName, Fees) VALUES (101, 'Data Science', 45000.00), (102, 'Computer Science', 50000.00), (103, 'BCA', 40000.00), (104, 'BBA', 35000.00);
```



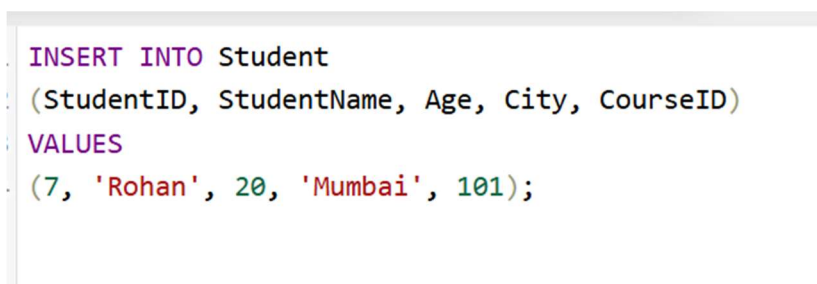
MySQL query execution interface showing a successful INSERT INTO statement. The query inserted 6 rows into the Student table. The interface includes a 'Show query box' button, a green success message, the SQL code, and links for 'Edit inline', 'Edit', and 'Create PHP code'.

```
INSERT INTO Student (StudentID, StudentName, Age, City, CourseID) VALUES (1, 'Amit', 20, 'Pune', 101), (2, 'Priya', 21, 'Delhi', 102), (3, 'Rahul', 19, 'Mumbai', 103), (4, 'Sneha', 22, 'Nashik', 104), (5, 'Karan', 20, 'Nagpur', 101), (6, 'Neha', 21, 'Aurangabad', 102);
```

Q3) Write a MySQL query to display all students enrolled in the BCA course using a subquery.

ANS

QUERY



MySQL query editor showing an INSERT INTO statement for the Student table. The query inserts a new student named Rohan, 20 years old, from Mumbai, enrolled in the BCA course (CourseID 103).

```
INSERT INTO Student (StudentID, StudentName, Age, City, CourseID) VALUES (7, 'Rohan', 20, 'Mumbai', 103);
```

OUTPUT

```
✓ 1 row inserted. (Query took 0.0897 seconds.)

INSERT INTO Student (StudentID, StudentName, Age, City, CourseID) VALUES (7, 'Rohan', 20, 'Mumbai', 101);

[ Edit inline ] [ Edit ] [ Create PHP code ]
```

## QUERY

```
1 SELECT * FROM Student;
```

## OUTPUT

| Extra options            |                  |           |             |                    |                                |                   |
|--------------------------|------------------|-----------|-------------|--------------------|--------------------------------|-------------------|
|                          |                  | Show all  |             | Number of rows: 20 | Filter rows: Search this table | Sort by key: none |
|                          |                  | StudentID | StudentName | Age                | City                           | CourseID          |
| <input type="checkbox"/> | Edit Copy Delete | 1         | Amit        | 20                 | Pune                           | 101               |
| <input type="checkbox"/> | Edit Copy Delete | 2         | Priya       | 21                 | Mumbai                         | 102               |
| <input type="checkbox"/> | Edit Copy Delete | 3         | Rahul       | 19                 | Mumbai                         | 103               |
| <input type="checkbox"/> | Edit Copy Delete | 4         | Sneha       | 22                 | Nashik                         | 104               |
| <input type="checkbox"/> | Edit Copy Delete | 5         | Karan       | 20                 | Nagpur                         | 101               |
| <input type="checkbox"/> | Edit Copy Delete | 6         | Neha        | 21                 | Aurangabad                     | 102               |
| <input type="checkbox"/> | Edit Copy Delete | 7         | Rohan       | 20                 | Mumbai                         | 101               |

Q4) Write a MySQL query to display each student with the overall average marks using a subquery in the SELECT clause.

ANS

## QUERY

```
1 SELECT * FROM Student
2 WHERE StudentID = 2;
```

## OUTPUT

| ←T→                                                                                         | StudentID | StudentName | Age | City   | CourseID |
|---------------------------------------------------------------------------------------------|-----------|-------------|-----|--------|----------|
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 2         | Priya       | 21  | Mumbai | 102      |

## QUERY

```

1 UPDATE Student
2 SET City = 'Mumbai'
3 WHERE StudentID = 2;

```

## OUTPUT

✓ 0 rows affected. (Query took 0.0007 seconds.)

```
UPDATE Student SET City = 'Mumbai' WHERE StudentID = 2;
```

## QUERY

```

1 SELECT * FROM Student
2 WHERE StudentID = 2;

```

## OUTPUT

| Extra options                                                                                                                                                                    |           |             |     |        |          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|-----|--------|----------|
| ←T→                                                                                                                                                                              | StudentID | StudentName | Age | City   | CourseID |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete                                                                                      | 2         | Priya       | 21  | Mumbai | 102      |
| <input type="checkbox"/> Check all    With selected: <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete <input type="checkbox"/> Export |           |             |     |        |          |

Q5) Write a MySQL query using a correlated subquery to display students who scored more than their own course average.

ANS

QUERY

```

1 SELECT * FROM Course
2 WHERE CourseID = 101;

```

## OUTPUT

Extra options

|                                             | CourseID | CourseName   | Fees     |
|---------------------------------------------|----------|--------------|----------|
| <input type="checkbox"/> Edit  Copy  Delete | 101      | Data Science | 47000.00 |

## QUERY

```

1 UPDATE Course
2 SET Fees = 47000
3 WHERE CourseID = 101;

```

## OUTPUT

✓ 0 rows affected. (Query took 0.0009 seconds.)

UPDATE Course SET Fees = 47000 WHERE CourseID = 101;

## QUERY

```

1 SELECT * FROM Course
2 WHERE CourseID = 101;

```

## OUTPUT

|                                             | CourseID | CourseName   | Fees     |
|---------------------------------------------|----------|--------------|----------|
| <input type="checkbox"/> Edit  Copy  Delete | 101      | Data Science | 47000.00 |

Q6) Write MySQL queries to display courses having students and courses having no students.

ANS

## QUERY

```
1 DELETE FROM Student
2 WHERE StudentID = 3;
```

## OUTPUT

✔ 0 rows deleted. (Query took 0.0011 seconds.)

```
DELETE FROM Student WHERE StudentID = 3;
```

|                          |                    | StudentID | StudentName | Age | City       | CourseID |
|--------------------------|--------------------|-----------|-------------|-----|------------|----------|
| <input type="checkbox"/> | Edit  Copy  Delete | 1         | Amit        | 20  | Pune       | 101      |
| <input type="checkbox"/> | Edit  Copy  Delete | 2         | Priya       | 21  | Mumbai     | 102      |
| <input type="checkbox"/> | Edit  Copy  Delete | 3         | Rahul       | 19  | Mumbai     | 103      |
| <input type="checkbox"/> | Edit  Copy  Delete | 4         | Sneha       | 22  | Nashik     | 104      |
| <input type="checkbox"/> | Edit  Copy  Delete | 5         | Karan       | 20  | Nagpur     | 101      |
| <input type="checkbox"/> | Edit  Copy  Delete | 6         | Neha        | 21  | Aurangabad | 102      |
| <input type="checkbox"/> | Edit  Copy  Delete | 7         | Rohan       | 20  | Mumbai     | 101      |